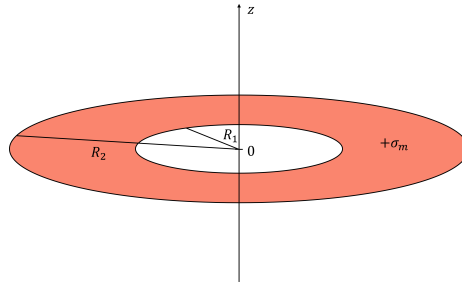
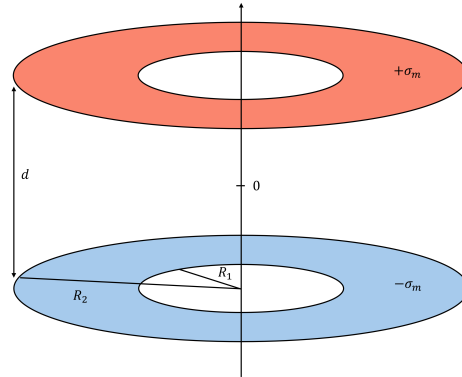


1 Vlakke oppervlakken

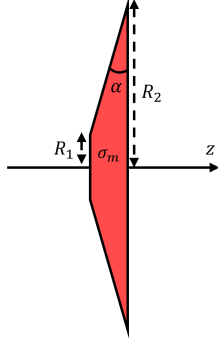


$$B_{z,ring}(z) = \frac{\mu_0 \sigma_m z}{2} \left(\frac{1}{\sqrt{z^2 + R_1^2}} - \frac{1}{\sqrt{z^2 + R_2^2}} \right) \quad (1)$$



$$B_z(z) = B_{z,ring}\left(z - \frac{d}{2}\right) - B_{z,ring}\left(z + \frac{d}{2}\right) \quad (2)$$

2 Gekantelde oppervlakken



$$h = (R_2 - R_1) \tan(\alpha)$$

$$k = \frac{R_2 - R_1}{h} = \cot(\alpha)$$

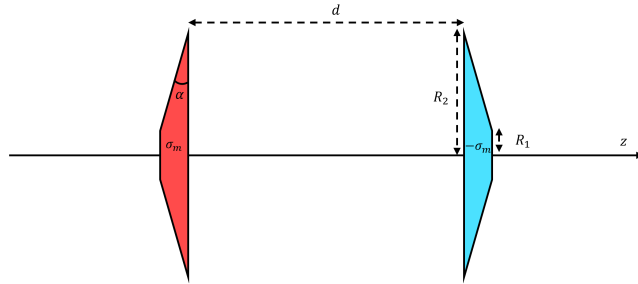
$$s = \sqrt{k^2 + 1} = \csc(\alpha)$$

$$B_{z,ring,\alpha}(z) = \frac{\|h\|}{h} \frac{\sigma_m \mu_0 s}{2} (T_1 + T_2 + T_3) \quad (3)$$

$$T_1 = \frac{kz - R_1}{s^2 \sqrt{R_1^2 + z^2}} \quad (4)$$

$$T_2 = \frac{2hk + R_1 - kz}{s^2 \sqrt{R_1^2 + h^2 s^2 + 2h(kR_1 - z) + z^2}} \quad (5)$$

$$T_3 = \frac{k}{s^3} \log \left[\frac{kR_1 - z + s \sqrt{R_1^2 + z^2}}{hs^2 + kR_1 - z + s \sqrt{2hkR_1 + R_1^2 + h^2 s^2 - 2hz + z^2}} \right] \quad (6)$$



$$B_z(z) = B_{z,ring,\alpha} \left(z + \frac{d + 2\|h\|}{2} \right) - B_{z,ring,-\alpha} \left(z - \frac{d + 2\|h\|}{2} \right) \quad (7)$$